

BHARGAV BIRE

Senior AI/ML Site Reliability Engineer | Infrastructure & DevOps

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PROFESSIONAL SUMMARY

- Senior AI/ML Site Reliability Engineer with 7+ years of experience across data engineering, ML delivery, and enterprise AI platform operations on Microsoft Azure and AWS
- Administer enterprise AI Gateway infrastructure built on LiteLLM Proxy, serving 10M+ monthly inference requests for 50+ application teams with model routing, rate limiting, load balancing, and automated failover
- Own reliability, monitoring, and observability for business-critical AI systems including LangSmith, AI-OrBit, and the AI Metrics Dashboard, sustaining a 99.9% uptime SLA through proactive incident detection
- Manage LLM deployments across Azure OpenAI, AI Foundry, and Amazon Bedrock, covering capacity planning, PTU allocation, token governance, and model version upgrades
- Operate self-hosted LLM tracing and evaluation infrastructure on AWS EKS and ECS, backed by Aurora PostgreSQL, ClickHouse, and ElastiCache Redis
- Architect cloud infrastructure with Terraform, Kubernetes (EKS, OpenShift), and Helm across AWS and Azure, including multi-AZ high availability and disaster recovery design
- Deliver FinOps chargeback and consumption reporting at 95% attribution accuracy, giving application teams visibility into token usage, spend, and budget consumption
- Engineer CI/CD pipelines on Jenkins and GitHub Actions, shipping 100+ production releases annually at a 98% deployment success rate
- Develop full-stack AI applications in Python, FastAPI, and React, including the Ask OrBie agentic assistant (Claude Agent SDK, MCP), chargeback analytics platform, Vanguard RCA system, and Model ComparEval
- Diagnose production issues using Splunk, Dynatrace APM, and distributed tracing, reducing mean time to resolution by 40%

TECHNICAL SKILLS

AI/ML Platforms	LangSmith, LiteLLM Proxy, LangChain, LangGraph, Azure OpenAI, AI Foundry, Amazon Bedrock, Claude Agent SDK, MCP, Anthropic API, OpenAI API, Vector Search
Cloud Platforms	AWS (EKS, ECS, Lambda, Aurora PostgreSQL, S3, Route 53, ALB/NLB, CloudWatch, EventBridge, Secrets Manager), Azure (AI Search, OpenAI, ExpressRoute, PTU)
Infrastructure & IaC	Terraform, Helm, Kubernetes, Docker, OpenShift, CloudFormation, Nginx, HAProxy, cert-manager, External Secrets Operator, SealedSecrets
CI/CD & DevOps	Jenkins, GitHub Actions, AWS CodePipeline, CodeBuild, ECR, Trivy, SonarQube, ArgoCD, GitOps, Blue-Green Deployments
Monitoring	Splunk, Dynatrace, CloudWatch, Grafana, Prometheus, ELK Stack, OpenTelemetry, Jaeger, KEDA, HPA, APM, Performance Testing
Databases & Storage	ClickHouse, PostgreSQL, Aurora, SQL Server, Redis, MongoDB, Vector Search, S3, Azure Blob Storage
Languages & Tools	Python, FastAPI, Pydantic, SQLAlchemy, React 18, Recharts, Node.js, SQL, Shell/Bash, Git, REST APIs, WebSocket, Okta, OAuth 2.0, Jira

PROFESSIONAL EXPERIENCE

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Cigna Healthcare / Evernorth, CT | June 2024 – Present

- Own monitoring, observability, and root-cause analysis for enterprise AI platforms (AI Gateway, AI-OrBit, LangSmith, AI Metrics Dashboard) using Splunk log analytics (50K+ daily events), Dynatrace traces, and cloud telemetry to detect and diagnose incidents against a sub-200ms response-time SLA
- Produce Daily Health Check (DHC) reports and validate operational KPIs (latency P50/P95/P99, error rates, throughput), correlating incidents across 15+ business-critical AI services for executive review
- Administer the enterprise AI Gateway and LiteLLM Proxy handling 10M+ monthly requests, configuring 100+ model routes, rate limits (1000 req/min), routing rules, and failover policies with zero-downtime migrations
- Run performance and load testing to validate throughput (10K+ req/min) and latency budgets, feeding capacity plans that raised sustained throughput by 30%
- Implemented self-hosted LangSmith on AWS EKS and ECS to trace 5M+ LLM calls monthly, backed by Aurora PostgreSQL (500GB+), a 6-node ElastiCache Redis cluster, and ClickHouse (10TB+)
- Provisioned AWS infrastructure through 50+ reusable Terraform modules (VPC, EKS with 20+ node groups, ECR, S3, Route 53, ALB/NLB, RDS Aurora) with multi-AZ high availability, KEDA autoscaling that absorbs 3x traffic spikes, and documented failover runbooks meeting a 15-minute recovery time objective
- Deployed 25+ Azure OpenAI and AI Foundry models (GPT-4o, GPT-4.1, GPT-5, O3) against 1M+ PTU of provisioned capacity, tuning allocation to reach 85% PTU utilization
- Architected the AI-OrBit platform as 12+ FastAPI microservices with a React front end and Redis caching, serving 50+ teams at 99.9% uptime
- Developed the Ask OrBie conversational assistant end to end on the Claude Agent SDK, with MCP tool routing, streaming WebSocket responses across 1000+ concurrent connections, and Okta M2M authentication, handling 50K+ conversations monthly
- Built custom MCP servers exposing enterprise capabilities (runbook search, incident lookup, report generation, governed SQL execution) plus the Ask OrBie React chat interface with token-by-token streaming, conversation history in Redis and SQL Server, feedback capture, and a shared prompt-template library
- Delivered the Vanguard alerting system with Webex integration, LLM-generated root-cause summaries, and queue-based processing of 1000+ alerts monthly, cutting MTTR from 45 to 27 minutes
- Created the Model ComparEval platform for side-by-side LLM evaluation across GPT-4, GPT-4.1, GPT-5, and O3, standardizing benchmarking and model-approval governance
- Established CI/CD on Jenkins and GitHub Actions with 10+ self-hosted runners, reaching a 98% deployment success rate across 100+ releases, with Trivy scanning on every ECR image
- Developed the chargeback engine in Python and FastAPI, aggregating LiteLLM, Azure OpenAI, and Bedrock token usage into per-team, per-model, and per-application attribution with configurable rate cards and monthly reconciliation at 95% accuracy
- Built the Chargeback Dashboard in React with Recharts for spend-trend analysis, model-level token breakdowns, and Okta-scoped cost-center drill-downs, fed by 6 scheduled AWS Lambda and EventBridge jobs that process 500K+ monthly transactions into S3 CSV and Excel exports emailed to 50+ cost-center owners
- Developed the GenAI ExpoLab analytics service, exposing model adoption, prompt volume, and usage trends through REST APIs consumed by leadership dashboards
- Hardened security and compliance for AI workloads with Okta SSO and M2M authentication, OAuth 2.0 scopes, AWS Secrets Manager, SealedSecrets, and HIPAA-aligned data governance controls

Data Engineer / Platform Engineer

Analytics Domain, India | January 2021 – February 2023

- Engineered ETL and ELT pipelines on AWS Glue, PySpark, and Apache Airflow processing 5+ TB daily (150TB+ monthly) at 99.5% data accuracy, plus Kafka streaming pipelines delivering sub-minute data freshness
- Architected the AWS foundation from the ground up in Terraform: VPC across 3 availability zones, NAT gateways, and security groups supporting 20+ services
- Built and operated EKS clusters running 50+ containerized workloads with autoscaling policies and Helm-based deployments, adding AWS App Mesh for service-to-service communication
- Deployed serverless pipelines on AWS Lambda, Step Functions, and EventBridge processing 10M+ events daily at 99.9% reliability
- Configured a 500TB+ S3 data lake with lifecycle, cross-region replication, and Glacier archival policies, and stood up RDS (PostgreSQL, MySQL) and Redshift warehouses with automated backups that cut report generation time by 65%
- Implemented CI/CD on AWS CodePipeline, CodeBuild, and GitHub Actions, containerizing 15+ services and reducing deployment time by 60%
- Developed CloudWatch dashboards and alarms across 100+ infrastructure metrics, plus Power BI and Tableau reporting used by 20+ business stakeholders
- Automated environment provisioning with reusable Terraform modules and Jenkins pipelines, shortening setup from two weeks to two days
- Provided L3 production support for 100+ pipelines on a 24x7 on-call rotation, running incident postmortems and runbook automation to hold SLA compliance at 99.5%

Data Engineer

Max Health Care, India | April 2018 – December 2020

- Designed ETL and ELT pipelines processing 2M+ patient records daily (60M+ monthly) on Azure Data Factory, PySpark, and Talend at 99.8% data integrity
- Built a healthcare data warehouse integrating HL7 and FHIR feeds, EMR extracts, and laboratory systems, and modeled patient-journey analytics adopted by care-coordination teams across 20+ departments
- Engineered real-time clinical dashboards streaming vital signs and lab results from 500+ connected devices, reducing critical-alert response time by 35%
- Automated clinical reporting pipelines generating 10,000+ reports monthly and removing 40% of manual reporting effort
- Implemented a data quality framework with automated validation rules, reaching 99.9% accuracy across patient demographic and billing datasets
- Developed Python automation and reusable connectors for EMR, PACS, and pharmacy systems, and documented data dictionaries and lineage for 50+ datasets
- Enforced HIPAA-aligned data governance and privacy controls, clearing 100% of internal audits over three years, and delivered Power BI dashboards used by 100+ clinical and operational staff
- Supported production operations on call, resolving 200+ pipeline incidents and authoring runbooks that reduced MTTR by 45%
- Tuned SQL Server and Azure Synapse workloads through indexing, partitioning, and stored-procedure optimization, improving query performance by 60%

EDUCATION

Master of Science in Business Analytics

University of New Haven, West Haven, Connecticut, USA | March 2024

Bachelor of Technology in Computer Science and Engineering

Gokaraju Rangaraju Institute of Engineering and Technology, Hyderabad, India | April 2019